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# Task1
s = "Python Programming Language"
print(s[7:18])
print(s[::-1])
print(s[1:2])
print(s[-1:-11:-1])
print("="*60)

# Task2
s = "Artificial Intelligence"
print(s[11:])
print(s[:10][::-1])
print(s[:10][::-1] + " " + s[11:][::-1])
print("="*60)

# Task3
lst = [5, 15, 25, 35, 45]
lst.insert(3, 30)
lst.remove(15)
lst.append(50)
lst[2] = 100
print(lst)
print("="*60)

# Task4
lst = ["Apple", "Banana", "Cherry"]
lst.insert(0, "Mango")
lst.insert(2, "Orange")
lst.remove("Apple")
lst.sort()
print(lst)
print("="*60)

# Task5
matrix = [
    [2,4,6],
    [8,10,12],
    [14,16,18]
]
print([row[0] for row in matrix])
print([matrix[i][i] for i in range(3)])
print(matrix[-1])
print("="*60)

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# Task6
matrix = [
    [1,2,3],
    [4,5,6]
]
matrix.append([7,8,9])
matrix.insert(0,[0,0,0])
print(matrix)
print("="*60)

# Task7
import json

data = [
    {"name": "A", "marks": 80},
    {"name": "B", "marks": 60}
]

with open("data.json","w") as f:
    json.dump(data,f)

with open("data.json","r") as f:
    d = json.load(f)

topper = max(d, key=lambda x: x["marks"])
print(topper)

d.append({"name": "C", "marks": 90})

with open("data.json","w") as f:
    json.dump(d,f)
print("="*60)

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# Task8
def check_email(e):
    if "@" in e and (e.endswith(".com") or e.endswith(".in")):
        return True
    return False

print(check_email("test@gmail.com"))
print("="*60)

# Task9
import random
num = random.randint(1,100)

for i in range(5):
    guess = int(input("Enter number: "))
    if guess == num:
        print("Correct")
        break
    elif guess > num:
        print("Too High")
    else:
        print("Too Low")
print("="*60)

# Task10
def primes(n):
    res = []
    for i in range(2,n+1):
        for j in range(2,i):
            if i%j==0:
                break
        else:
            res.append(i)
    print(res)
    print(len(res))
    print(max(res))

primes(50)
print("="*60)

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# Task11
contacts = {}

def add(name,num):
    contacts[name] = num

def search(name):
    print(contacts.get(name, "Not Found"))

def delete(name):
    contacts.pop(name, None)

def display():
    print(contacts)

add("Ali", 123)
add("Sara", 456)
search("Ali")
delete("Sara")
display()
print("="*60)

# Task12
import cv2
img = cv2.imread("car.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
edges = cv2.Canny(gray, 100, 200)
resized = cv2.resize(img, (200, 200))

cv2.imshow("Gray", gray)
cv2.imshow("Edges", edges)
cv2.imshow("Resized", resized)
cv2.waitKey(0)
cv2.destroyAllWindows()
print("="*60)
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# Task13
import threading

def numbers():
    for i in range(5):
        print(i)

def alphabets():
    for i in "ABCDE":
        print(i)

t1 = threading.Thread(target=numbers)
t2 = threading.Thread(target=alphabets)

t1.start()
t2.start()
print("="*60)

# Task14
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
import joblib
import tkinter as tk

df = pd.read_csv("iris.csv")

x = df.drop(columns=["Id", "Species"])
y = df["Species"]

x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2)

model = DecisionTreeClassifier()
model.fit(x_train, y_train)

joblib.dump(model, "model.pkl")
print("Model trained and saved")

def predict():
    vals = [
        float(e1.get()),
        float(e2.get()),
        float(e3.get()),
        float(e4.get())
    ]
    res = model.predict([vals])
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        res = model.predict([vals])
        result.config(text=res)

root = tk.Tk()
root.title("Prediction App")
root.geometry("300x350")

tk.Label(root, text="Iris Prediction", font=("Arial", 14)).pack(pady=10)

tk.Label(root, text="Sepal length").pack()
e1=tk.Entry(root)
e1.pack(pady=5)

tk.Label(root, text="Sepal width").pack()
e2=tk.Entry(root)
e2.pack(pady=5)

tk.Label(root, text="Petal length").pack()
e3=tk.Entry(root)
e3.pack(pady=5)

tk.Label(root, text="Petal width").pack()
e4=tk.Entry(root)
e4.pack(pady=5)

tk.Button(root, text="Predict", command=predict).pack(pady=10)

result=tk.Label(root, text="")
result.pack(pady=10)

root.mainloop()
print ("="*60)

# Task15
import librosa as lb
import matplotlib.pyplot as plt
import librosa.display

y, sr = lb.load("D:\\\\Internship\\\\Day21\\\\game_over.wav", sr=None)

print("Sampling Rate:", sr)

plt.figure(figsize=(10, 4))
lb.display.waveshow(y, sr=sr)

plt.title("Waveform")

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plt.title("Waveform")
plt.xlabel("Time")
plt.ylabel("Amplitude")

plt.show()
print("="*60)

# Task16
import cv2
import numpy as np

img = cv2.imread("car.jpg")

gray = cv2.cvtColor(img,cv2.COLOR_BGR2GRAY)
print("Grayscale applied")

blur = cv2.GaussianBlur(img, (7,7),0)
print("Blur applied")

kernel = np.array([[0,-1,0],[-1,5,-1],[0,-1,0]])
sharp = cv2.filter2D(img,-1,kernel)
print("Sharpen applied")

cv2.imshow("Original",img)
cv2.imshow("Gray",gray)
cv2.imshow("Blur",blur)
cv2.imshow("Sharpen",sharp)

cv2.waitKey(0)
cv2.destroyAllWindows()
print("="*60)

# Task17
class Employee:
    def __init__(self,n,i,s):
        self.name=n
        self.id=i
        self.salary=s

    def inc(self,amt):
        self.salary+=amt

    def display(self):
        print(self.name,self.id,self.salary)

    def annual(self):

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        def annual(self):
            print(self.salary*12)

e = Employee("Ali",1,20000)
e.inc(5000)
e.display()
e.annual()
print("="*60)

# Task18 & 19
class Vehicle:
    def __init__(self,name,rate):
        self.name=name
        self.rate=rate

class Car(Vehicle):
    pass

class Bike(Vehicle):
    pass

v = Car("Car",100)
print(v.name,v.rate)
print("="*60)

# Task20
class Patient:
    def __init__(self,name):
        self.name=name

class Doctor:
    def __init__(self,name):
        self.name=name

class Appointment:
    def __init__(self,p,d):
        self.p=p
        self.d=d

a = Appointment("Ali", "Dr.X")
print(a.p,a.d)
print("="*60)

# Task21
class Order:
    def __init__(self):

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|  
# Task21  
class Order:  
    def __init__(self):  
        self.items=[]  
  
    def add(self,item,price):  
        self.items.append(price)  
  
    def total(self):  
        return sum(self.items)  
  
o = Order()  
o.add("Burger",100)  
o.add("Pizza",200)  
print(o.total())
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===== RESTART: C:/Users/krsb/OneDrive/Desktop/internship/DA
Programming
egaugnaL gnimmargorP nohtyP
yhnPormigLnug
egaugnaL g
=====
Intelligence
laicifitrA
laicifitrA ecnegilletnI
=====
[5, 25, 100, 35, 45, 50]
=====
['Banana', 'Cherry', 'Mango', 'Orange']
=====
[2, 8, 14]
[2, 10, 18]
[14, 16, 18]
=====
[[0, 0, 0], [1, 2, 3], [4, 5, 6], [7, 8, 9]]
=====
{'name': 'A', 'marks': 80}
=====
True
=====
Enter number: 52
Too Low
Enter number: 5699
Too High
Enter number: 566
Too High
Enter number: 210
Too High
Enter number: 3
Too Low
=====
[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
15
47
=====
123
{'Ali': 123}
=====
```

